

In this paper, we denote that  $\alpha_m: \mathbb{N} \rightarrow \mathbb{R}$ , ( $m \in \mathbb{N}$ ).

Def. A set of sequences that converges to zero

$$C_0(\mathbb{N}) \stackrel{\text{def}}{=} \{ \alpha: \mathbb{N} \rightarrow \mathbb{R} \mid \lim_{n \rightarrow \infty} \alpha(n) = 0 \}$$

And define the  $\ell^1$ -space

$$\ell^1(\mathbb{N}) \stackrel{\text{def}}{=} \{ \alpha \in C_0(\mathbb{N}) \mid \text{the series } \sum_{n=1}^{\infty} \alpha(n) \text{ converges absolutely} \}$$

It is a normed space, with a norm

$$\| \alpha \|_1 = \sum_{n=1}^{\infty} | \alpha(n) |.$$

Thm. The normed space  $\ell^1(\mathbb{N})$  is complete metric space.

Proof. Let  $\{ \alpha_m : \mathbb{N} \rightarrow \mathbb{R} \}$  be a Cauchy sequence. Then, for each  $n \in \mathbb{N}$ ,

$$| \alpha_m(n) - \alpha_k(n) | \leq \sum_{i=1}^{\infty} | \alpha_m(i) - \alpha_k(i) | = \| \alpha_m - \alpha_k \|_1, \quad \text{for any } m, k \in \mathbb{N}.$$

So, since  $\{ \alpha_m \}$  is Cauchy sequence,  $\{ \alpha_m(n) \}$  is also Cauchy sequence for each  $n \in \mathbb{N}$ .

The  $\mathbb{R}$  is complete, define  $\alpha(n) = \lim_{m \rightarrow \infty} \alpha_m(n)$  for each  $n \in \mathbb{N}$ .

To show that  $\alpha \in \ell^1(\mathbb{N})$  and  $\| \alpha_m - \alpha \|_1 \rightarrow 0$  as  $m \rightarrow \infty$ , let  $\varepsilon > 0$  be given.

Since  $\{ \alpha_m \}$  is Cauchy sequence, there exists  $N \in \mathbb{N}$  such that

$$m, k \geq N \Rightarrow \| \alpha_m - \alpha_k \|_1 < \varepsilon.$$

Then, for each  $n \in \mathbb{N}$ ,

$$m, k \geq N \Rightarrow \sum_{i=1}^n | \alpha_m(i) - \alpha_k(i) | \leq \| \alpha_m - \alpha_k \|_1 < \varepsilon.$$

Letting  $k \rightarrow \infty$ , we obtain that, for all  $n \in \mathbb{N}$

$$m \geq N \Rightarrow \sum_{i=1}^n | \alpha_m(i) - \alpha(i) | \leq \varepsilon.$$

So  $\| \alpha_m - \alpha \|_1 \rightarrow 0$  as  $m \rightarrow \infty$ .

In particular, for each  $n \in \mathbb{N}$ ,

$$\sum_{i=1}^n | \alpha(i) | \leq \sum_{i=1}^n | \alpha_N(i) | + \sum_{i=1}^n | \alpha(i) - \alpha_N(i) | \leq \| \alpha_N \|_1 + \varepsilon, \quad \text{so } \alpha \in \ell^1(\mathbb{N}).$$

□

